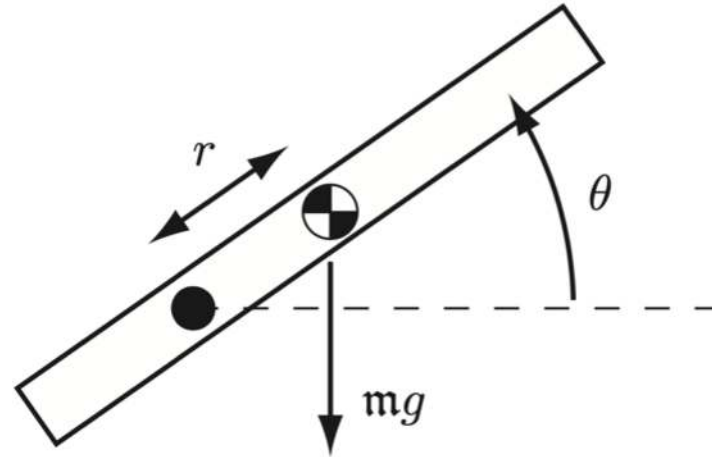


Chapter 2	Configuration Space
Chapter 3	Rigid-Body Motions
Chapter 4	Forward Kinematics
Chapter 5	Velocity Kinematics and Statics
Chapter 6	Inverse Kinematics
Chapter 7	Kinematics of Closed Chains
Chapter 8	Dynamics of Open Chains
Chapter 9	Trajectory Generation
Chapter 10	Motion Planning
Chapter 11	Robot Control
	11.1 Control System Overview
	11.2 Error Dynamics
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Chapter 12	Grasping and Manipulation
Chapter 13	Wheeled Mobile Robots

Important concepts, symbols, and equations

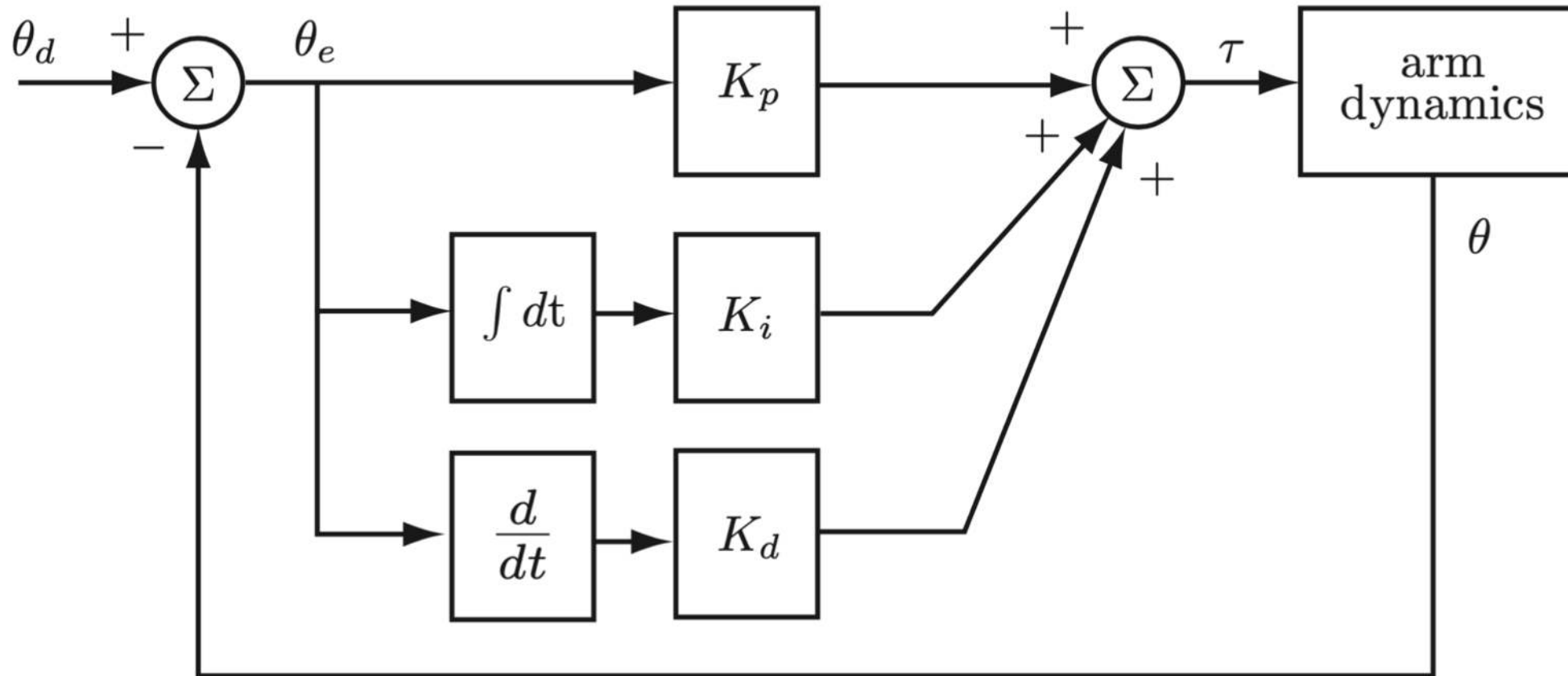


$$\tau = M\ddot{\theta} + \underbrace{mgr \cos \theta + b\dot{\theta}}_{h(\theta, \dot{\theta})}$$
$$\tau = M\ddot{\theta} + h(\theta, \dot{\theta})$$

Proportional-Integral-Derivative (PID) control

$$\tau = K_p \theta_e + K_i \int \theta_e(t) dt + K_d \dot{\theta}_e$$

Important concepts, symbols, and equations (cont.)



Important concepts, symbols, and equations (cont.)

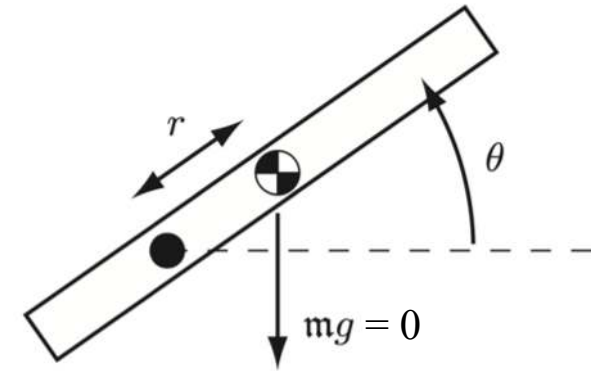
Setpoint PD control, $g = 0$

$$\tau = K_p \theta_e + \overset{0}{\cancel{K_i}} \int \theta_e(t) dt + K_d \dot{\theta}_e$$

$$\tau = M\ddot{\theta} + \overset{0}{\cancel{mgr}} \cos \theta + b\dot{\theta}$$

$$\dot{\theta}_d = \ddot{\theta}_d = 0$$

$$M\ddot{\theta} + b\dot{\theta} = K_p \theta_e + K_d \dot{\theta}_e$$



$$\ddot{\theta}_e + \frac{b + K_d}{M} \dot{\theta}_e + \frac{K_p}{M} \theta_e = 0$$

$$\zeta = \frac{b + K_d}{2\sqrt{K_p M}} \quad \omega_n = \sqrt{\frac{K_p}{M}}$$

For what gains
are the error
dynamics stable?

$$K_d > -b$$
$$K_p > 0$$

Important concepts, symbols, and equations (cont.)

Setpoint PD control, $g \neq 0$

$$\tau = K_p \theta_e + \overset{0}{\cancel{K_i}} \int \theta_e(t) dt + K_d \dot{\theta}_e$$

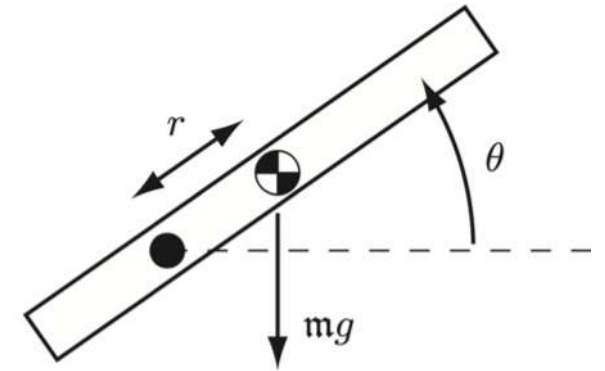
$$\tau = M\ddot{\theta} + mgr \cos \theta + b\dot{\theta}$$

$$\dot{\theta}_d = \ddot{\theta}_d = 0$$

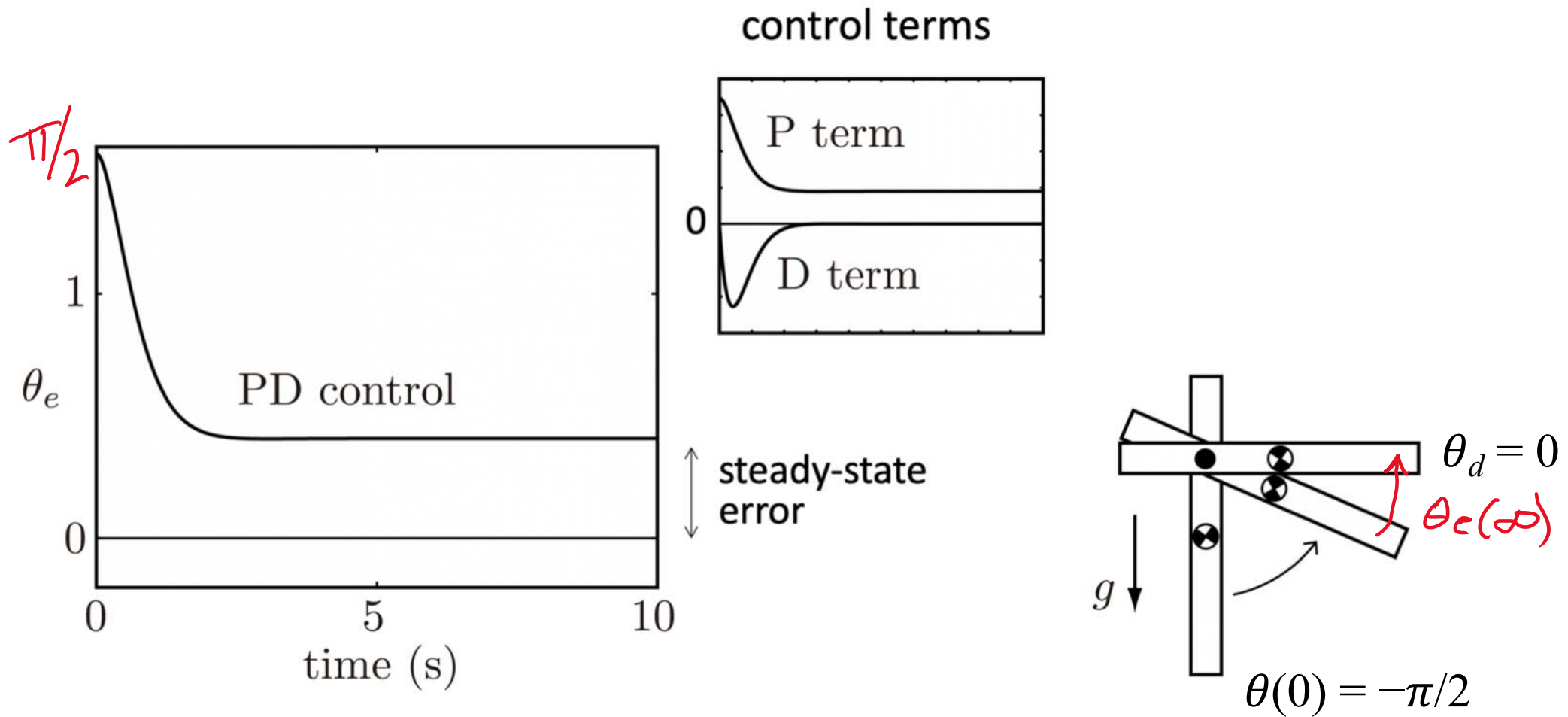
$$M\ddot{\theta}_e + (b + K_d)\dot{\theta}_e + K_p\theta_e = mgr \cos \theta$$

$$\theta_e(t \rightarrow \infty) = \frac{mgr \cos \theta}{K_p}$$

Nonhomogeneous.
What is the steady-state error?



Important concepts, symbols, and equations (cont.)



Important concepts, symbols, and equations (cont.)

Setpoint PID control, $g \neq 0$

$$\tau = K_p \theta_e + K_i \int \theta_e(t) dt + K_d \dot{\theta}_e$$

$$\tau = M\ddot{\theta} + \text{mgr} \cos \theta + b\dot{\theta}$$

$$\dot{\theta}_d = \ddot{\theta}_d = 0$$

τ_{dist}

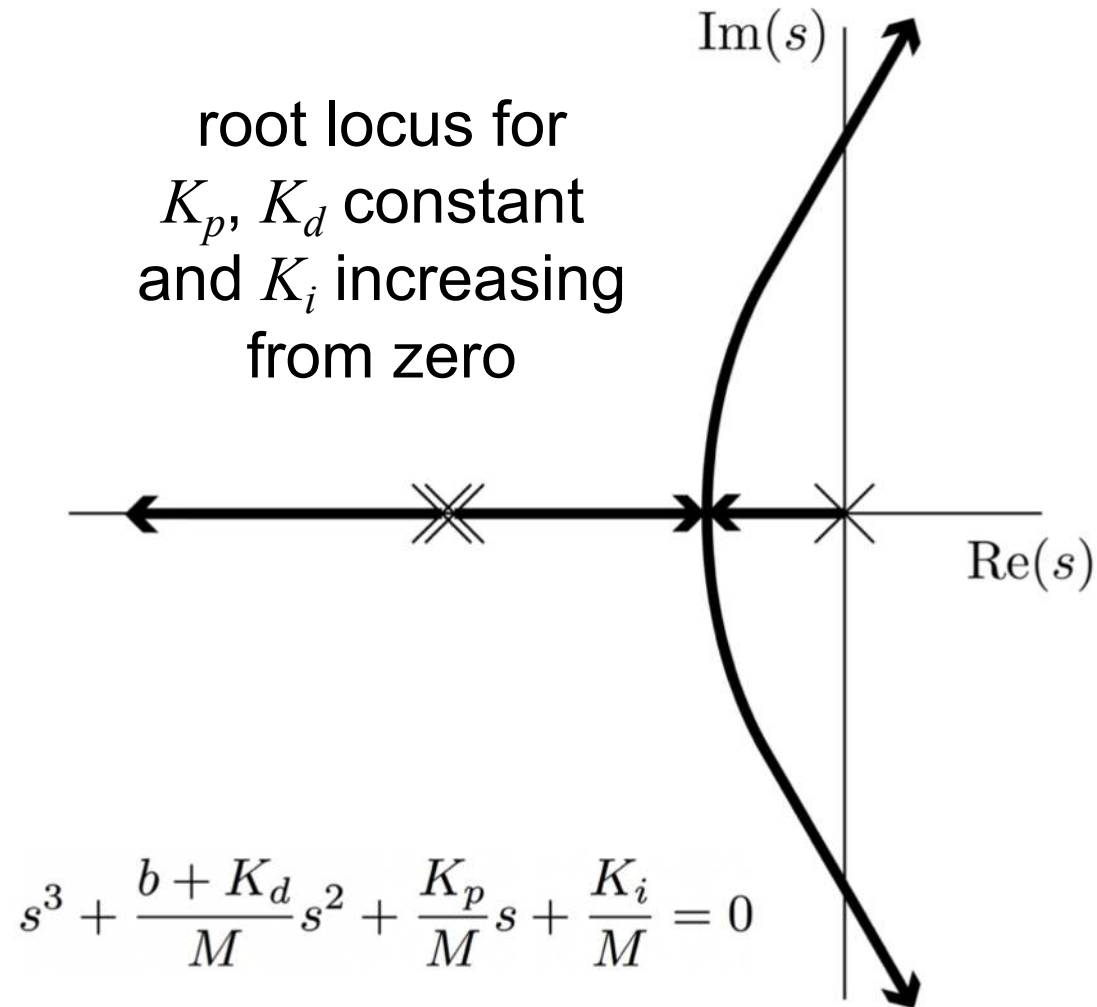
$\rightarrow$ because $\text{mgr} \cos \theta$ is nearly constant near steady state

$$M\ddot{\theta}_e + (b + K_d)\dot{\theta}_e + K_p\theta_e + K_i \int \theta_e(t) dt = \tau_{\text{dist}}$$

$$M\theta_e^{(3)} + (b + K_d)\ddot{\theta}_e + K_p\dot{\theta}_e + K_i\theta_e = 0$$

$$s^3 + \frac{b + K_d}{M}s^2 + \frac{K_p}{M}s + \frac{K_i}{M} = 0$$

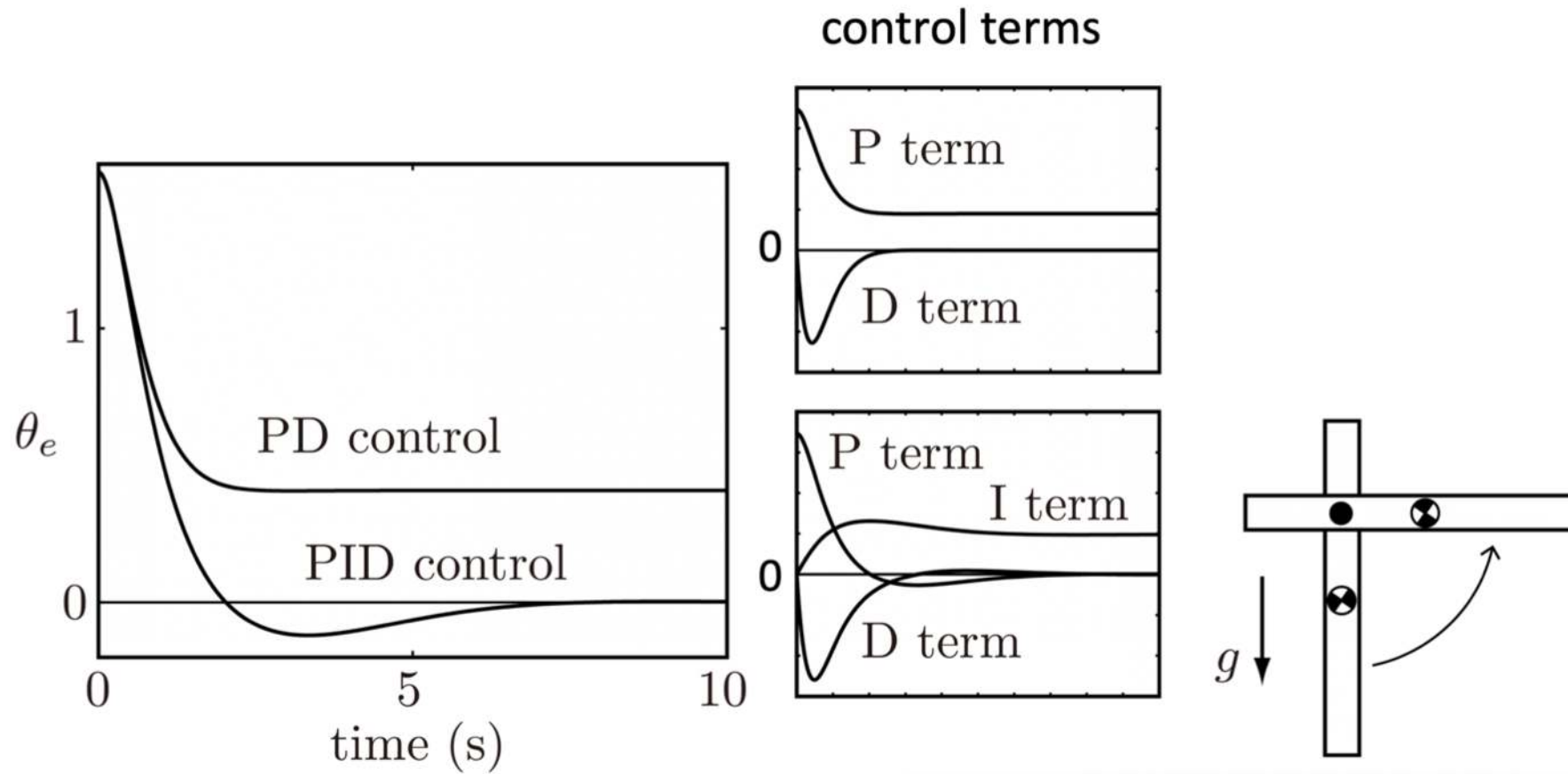
Important concepts, symbols, and equations (cont.)



$$\begin{aligned} K_d &> -b \\ K_p &> 0 \\ \frac{(b + K_d)K_p}{M} &> K_i > 0 \end{aligned}$$

K_i improves steady-state response but can worsen the transient response.

Important concepts, symbols, and equations (cont.)



Important concepts, symbols, and equations (cont.)

What about tracking general trajectories, not just setpoint control?

$$\tau = M \left(\overbrace{\ddot{\theta}_d + K_p \theta_e + K_i \int \theta_e dt + K_d \dot{\theta}_e}^{\text{commanded } \ddot{\theta}} \right) + h(\theta, \dot{\theta})$$

ff *fb*

$$\ddot{\theta}_e = \ddot{\theta}_d - \ddot{\theta}$$

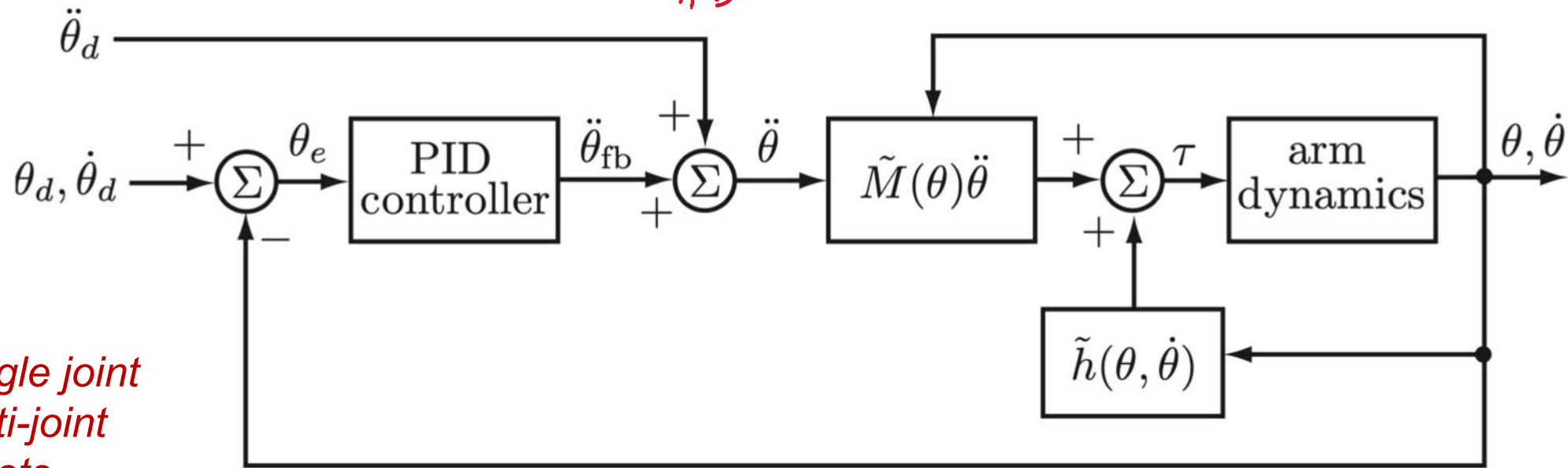
$$\ddot{\theta}_e = -K_d \dot{\theta}_e - K_p \theta_e - K_i \int \theta_e dt$$

$$\theta_e^{(3)} + K_d \ddot{\theta}_e + K_p \dot{\theta}_e + K_i \theta_e = 0$$

Important concepts, symbols, and equations (cont.)

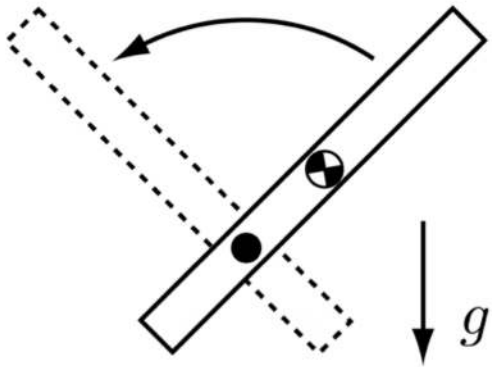
Computed torque control (feedback linearization)

$$\tau = \underbrace{\tilde{M}(\theta)}_{\text{model}} \left(\underbrace{\ddot{\theta}_d}_{\text{ff}} + \underbrace{K_p \theta_e + K_i \int \theta_e(t) dt + K_d \dot{\theta}_e}_{\text{fb}} \right) + \underbrace{\tilde{h}(\theta, \dot{\theta})}_{\text{model}}$$

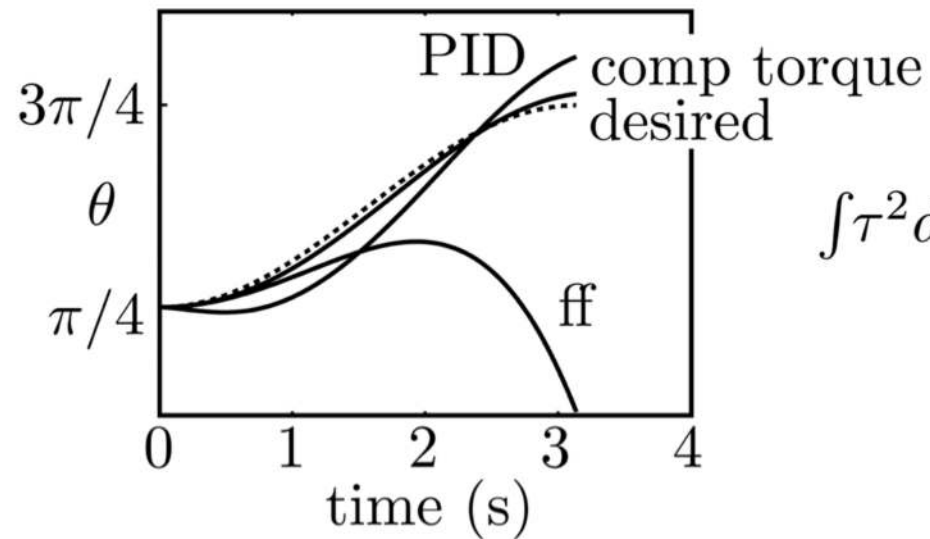


*for a single joint
or multi-joint
robots*

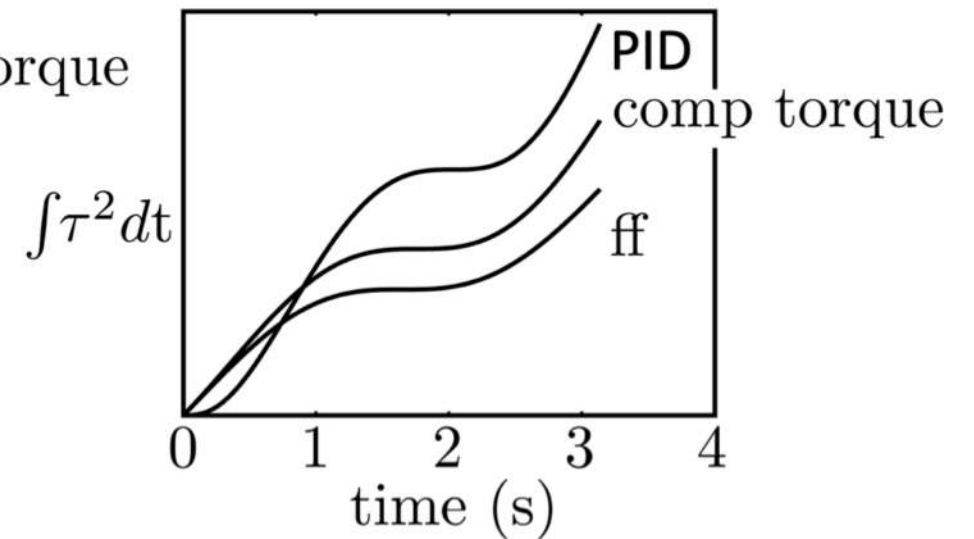
Important concepts, symbols, and equations (cont.)



assume the
model has some
error



better tracking
than ff or PID



less "effort" than PID

Important concepts, symbols, and equations (cont.)

Task-space computed torque control

$$\mathcal{F}_b = \Lambda(\theta)\dot{\mathcal{V}}_b + \eta(\theta, \mathcal{V}_b)$$

dynamic model: $\{\tilde{\Lambda}, \tilde{\eta}\}$

$$\tau = \tilde{M}(\theta) \left(\ddot{\theta}_d + K_p \theta_e + K_i \int \theta_e dt + K_d \dot{\theta}_e \right) + \tilde{h}(\theta, \dot{\theta})$$

$$\mathcal{F}_b = \tilde{\Lambda}(\theta) \left(\dot{\mathcal{V}}_d + K_p X_e + K_i \int X_e dt + K_d \mathcal{V}_e \right) + \tilde{\eta}(\theta, \mathcal{V}_b)$$

$$[X_e] = \log(X^{-1}X_d)$$

$$\mathcal{V}_e = [\text{Ad}_{X^{-1}X_d}] \mathcal{V}_d - \mathcal{V}_b$$

$$\tau = J_b^T(\theta) \mathcal{F}_b$$

What if your dynamic model is poor?

comp torque is appropriate if model is "reasonable" or better, not if it's poor

The characteristic equation of the error dynamics are

$$s^5 + 2s^4 + s^3 + 2s^2 + 4s + 2 = 0$$

Write the error dynamics in the form $\dot{x} = Ax$. Determine if the system is stable. (Use any software you want.)

$$x_1 = \theta_e$$

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = x_3$$

$$\dot{x}_3 = x_4$$

$$\dot{x}_4 = x_5$$

$$\dot{x}_5 = -2x_1 - 4x_2 - 2x_3 - x_4 - 2x_5$$

$$\text{eig}(A) = -1, -1, -1.26, \\ 0.68 \pm 1.09j$$

unstable due to
positive real component

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ -2 & -4 & -2 & -1 & -2 \end{bmatrix}$$